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Implications of new palaeomagnetic results from the Carboniferous System of Britain

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SUMMARY

The Carboniferous Limestone of the Craven Basin (northern England) gives a mean pole position of 40°N, 147°E which confirms the palaeolatitude of Britain relative to other continents at this time. A polarity scale has been constructed for the lower Carboniferous using the Avonian zonal scheme as a basis. The geomagnetic field was mostly reversed in lower Carboniferous times although brief periods of normal polarity were not uncommon, particularly in mid-Viséan times.

The evidence for a world-wide polar transition during the Carboniferous has been assessed and found to rest critically on the original evidence from Australia and Africa. The newer European and North American data are best explained in terms of a slow average motion of these two continents relative to the palaeomagnetic pole. They are shown to be mainly in low latitudes but to be rotating slowly clockwise with their southern edges in essentially equatorial locations for most of the Carboniferous.

MANY OF THE PALAEOMAGNETIC INVESTIGATIONS of Carboniferous rocks undertaken in Britain in the late 1950s were of stratigraphically restricted igneous rocks from N. England and central Scotland. Many of these observations have now been supplanted by more detailed analyses in which low coercivity components have been removed. Also, many of the new data presented here are from limestones, which have hitherto been regarded as being too weakly magnetic for proper palaeomagnetic analysis. A wide variety of Carboniferous limestone facies have been studied and shown to have magnetic properties which date to the time of their deposition. In particular, argillaceous, bituminous limestones are shown to be especially suitable for palaeomagnetic study. These occur widely throughout the lower Carboniferous and enable a rigorous palaeomagnetic-stratigraphical survey of this period for the first time.

Geology of the sampling area

The Carboniferous Limestone Series (Dinantian) of northern England was deposited in structurally contrasting 'blocks' and 'basins'. The blocks were relatively stable areas which underwent only gentle folding in post-Carboniferous times and were not submerged until after the start of the Viséan. In contrast, sedimentation in the basins started earlier, certainly in the Tournaisian (Johnson & Marshall 1971), and subsequent rapid subsidence was balanced by sedimentation so that great thicknesses of shallow water limestones and shales accumulated.

The Askrigg Block is separated from the Craven Basin by the Craven Fault system which includes the North-, Mid-, and South-Craven faults. These faults, particularly the Mid-Craven fault, were probably in existence in lower Carboniferous times and mark the hinge zone of the Craven Basin subsidence. The sediments of

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the Askrigg Block include the Great Scar Limestone and the overlying Yoredale Series. The Great Scar Limestone is about 200 m thick and consists mostly of massive, grey calcarenites of shallow marine origin. It ranges in age from C_2–D_1 of which usually just over 100 m are found to belong to the D_1 zone (Garwood & Goodyear 1924). The overlying Yoredales are about 300 m thick and consist of cyclothems, usually 15–30 m in thickness. Ideal cyclothems consist of a marine limestone followed by calcareous shales, ferruginous shales, sandstones, ganister and a thin coal deposited in deltaic environments.

In the Craven Basin a thick, mainly argillaceous, sequence accumulated, in places reaching *c.* 2000 m. The succession in the Clitheroe area is representative and its stratigraphy has been revised recently by Miller & Grayson (1972). The Chatburn Limestone Group is a series of dark, shaly bituminous limestones up to 300 m thick and of C_1–C_2 age. Overlying the Chatburn Limestone is the Worston Shale Group which contains at its base the Clitheroe Limestone Complex. This is a formation dominated by pure, fine grained limestones which at the present time form a series of distinct knolls, often with quaquaversal dip. The complex has a maximum thickness of above 800 m and is of C_2S_1 age.

Above the Clitheroe Limestone Complex are the Cephalopod Shales, about 580 m of thin limestones, shales and mudstones, probably of S_1 age. These are overlain by the *Bollandoceras hodderense* Beds, alternating thin micritic limestones and shales, up to 16 m thick and of S_2 age. Shales and thin limestones usually separate the *B. hodderense* Beds from the Pendleside Limestone whose top marks the top of the Worston Shale Group. The Pendleside Limestone has a maximum thickness of about 95 m and consists of alternating thin, dark grey, fine grained limestones and calcareous shales of D_1–D_2 age.

Sampling and measurements

Fifty-five oriented hand samples of limestones were collected from various localities in the Askrigg Block and Craven Basin areas. The sampled units include the Great Scar Limestone, Chatburn Limestone, Pendleside Limestone, *B. hodderense* Beds and the bank, flank, interbank and biosparite facies in the Clitheroe Limestone Complex. The location of the sites is shown on Fig. 1. Sixteen of the sites, all from the Ravenstonedale Basin, were found to be highly unstable and have not been included in the analysis. A further twenty-two hand samples were collected from Burrington Coombe and the Avon Gorge in the SW. Carboniferous province. However the magnetisation of these samples was found to be associated with secondary haematite and to post-date the Permo-Carboniferous folding (fold test) which affects the area so they also have been omitted from the analysis. A wide range of limestone lithologies was collected in order to assess their suitability for palaeomagnetic work. Their ages (in terms of the Avonian zonal scheme) range from C_1–C_2 (Chatburn Limestone) to D_1–D_2 (Pendleside Limestone). In the laboratory an optimum number of six cores was drilled perpendicular to the bedding in each sample. At least one specimen cylinder, 2·5 cm high, was cut from the base of each core.

The intensity, direction and homogeneity of remanence in each specimen (184 in

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all) were measured using a computerized 'Digico' balanced fluxgate, spinner magnetometer operated at 7 Hz (Molyneux 1971). The initial natural remanent magnetization (NRM) ranged from 0·03 to 7·43 $\times$ 10^{-6} gauss. The most weakly magnetized specimens were found to be the pure, pale-coloured limestones such as those from the reef-knolls (Salthill Bank Beds) in the Clitheroe Limestone Complex and the *B. hodderense* Beds. In contrast, the dark grey, fine-grained argillaceous limestones like the Pendleside Limestone and Chatburn Limestone were the most strongly magnetized (Tables 1, 3).

The directions of remanence for samples from the Craven Basin and Askrigg Block are shown in Figs. 2, 3. The Chatburn Limestone specimens are well grouped with a very shallow SSW inclination. Specimens from the Clitheroe Limestone Complex and *B. hodderense* Beds show a similar SSW declination but are scattered and have a moderately steep, positive (downward) inclination. Specimens from

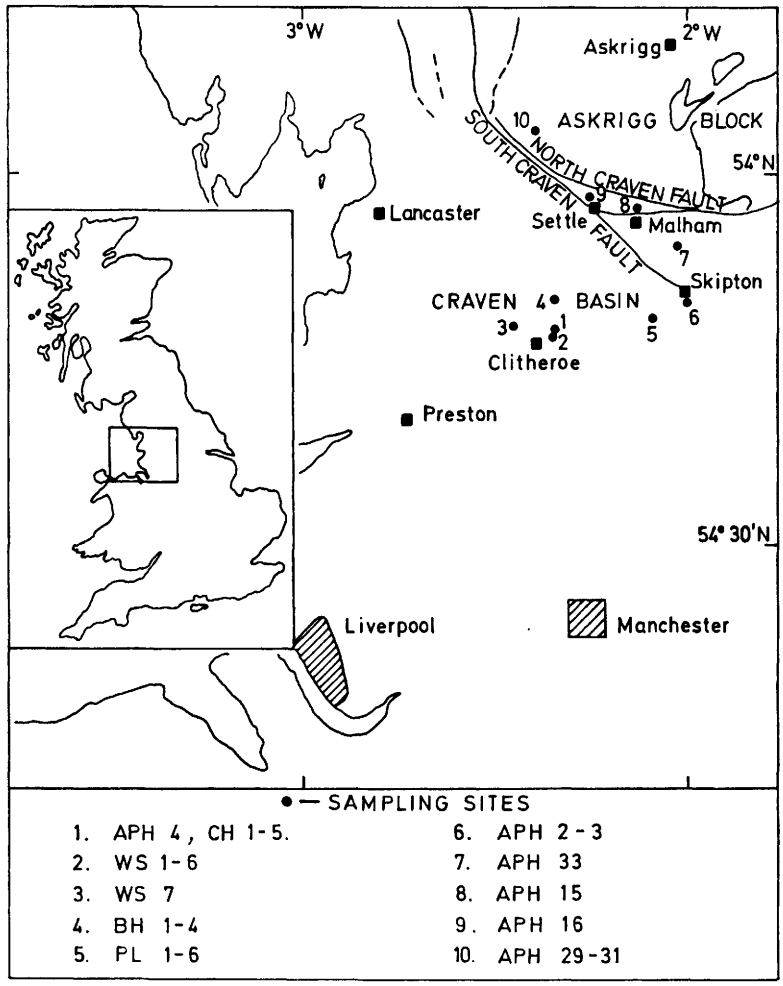


FIG. 1. Location map of sampling sites in the Carboniferous Limestone of the Craven Basin and Askrigg Block areas of northern England.

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two sites have NNE declination and variable steep positive dips. The specimens from the Pendleside Limestone are well grouped with a similar SSW declination and a negative (upward) inclination of *c.* 30°. The directions of remanence from the Askrigg Block (Great Scar Limestone) are scattered in the SW and NE quadrants.

One pilot sample was selected from each site and subjected to incremental thermal demagnetization, the remanence and low field susceptibility being measured after each temperature increment. Examples of the behaviour of pilot specimens in partial thermal demagnetization, including directions and intensities of remanence

TABLE 1: *Site mean directions and intensities of NRM for the Craven Basin. N refers to the number of specimens in each hand sample, k is the Fisher's (1953) precision parameter and* α_{95} *is the 95% cone of confidence for the mean.*

		Initial					200°C				
Site	N	Dec.	Inc.	Int.(x10^{-6} gauss)	k	α_{95}	Dec.	Inc.	Int.(x10^{-6} gauss	k	α_{95}
Pendleside Lst. (D_1-D_2)											
PL6	6	213	-24	1.15	476	3	217	-34	1.28	1247	2
PL5	6	189	-14	1.57	136	6	194	-26	1.89	223	4
PL4	4	216	-31	6.10	681	3	220	-36	6.85	552	4
PL3	4	220	-16	1.19	2	85	223	-38	1.31	2	77
PL2	6	217	-31	2.30	550	3	221	-34	2.53	944	2
PL1	4	215	-2	1.32	133	8	222	-19	1.29	364	5
Cracoe Reef Lst. (D_1)											
APH33	7	0	+24	0.05	11	19	352	+18	0.05	6	27
APH2	3	37	+43	1.19	13530	1	30	+35	1.20	270	7
APH3	3	48	+46	0.57	576	5	44	+35	0.54	188	9
B. hodderense Beds (S_2)											
BH4	6	245	+39	0.12	4	36	202	-5	0.14	84	7
BH3	5	193	+25	0.29	36	13	183	-3	0.34	939	2
BH2	4	213	+34	0.39	23	19	199	-2	0.26	187	7
BH1	6	199	+30	0.40	56	9	184	-2	0.26	378	3
Reef-knolls-Salthill (C_2S_1)											
APH6	6	198	+17	1.91	30	12	197	-9	1.11	100	7
APH7	6	184	+53	2.30	8	26	21	+59	1.00	113	6
WS1	2	200	+26	0.49	1164	7	193	-8	0.56	1016	8
WS2	6	191	+4	0.66	144	6	193	-20	0.69	831	2
WS3	6	181	+49	0.61	39	11	191	+5	0.52	741	2
WS4	6	8	+27	0.41	78	8	9	+1	0.49	252	4
WS5	6	211	+19	0.63	32	12	206	-6	0.45	37	11
WS6	4	199	+8	0.42	95	9	206	-22	0.50	524	4
Reef-knolls Dunnow (C_2-S_1)											
WS7	6	182	+32	0.10	22	15	206	+14	0.10	38	11
Chatburn Lst.(C_1C_2)											
CH1	3	216	+5	0.78	9	43	208	-13	0.83	10	40
CH2	2	211	+15	0.76	265	15	211	-20	0.88	1651	6
CH3	4	207	-2	0.41	123	8	212	-23	0.52	248	6
CH4	5	203	+1	0.31	73	9	201	-22	0.36	279	5
CH5	5	206	+12	0.28	58	10	198	-24	0.30	156	6
APH4	5	205	+4	1.07	567	5	205	-15	0.98	335	4

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and susceptibilities are shown in Fig. 4. Most of the pilot specimens from the Craven Basin showed good stability (Table 2). After a large initial change in the inclination from positive to negative (the removal of a viscous component near the present earth's field) most of the specimens took up a position with a S–SW declination and a shallow negative inclination (Fig. 4a). Specimens from the Pendleside Limestone showed little change from their initial directions and were highly stable. The most stable directions of two pilot samples, both from reef-knoll limestones, have NNW declinations and positive inclinations.

The pilot samples from the Great Scar Limestone were all weakly magnetized pure limestones. They were mostly unstable with stability indices (Tarling & Symons 1967) less than 2 (Table 2, Fig. 4b). In contrast to the Craven Basin specimens, most of the more stable pilot samples have a NE–NNE declination and a

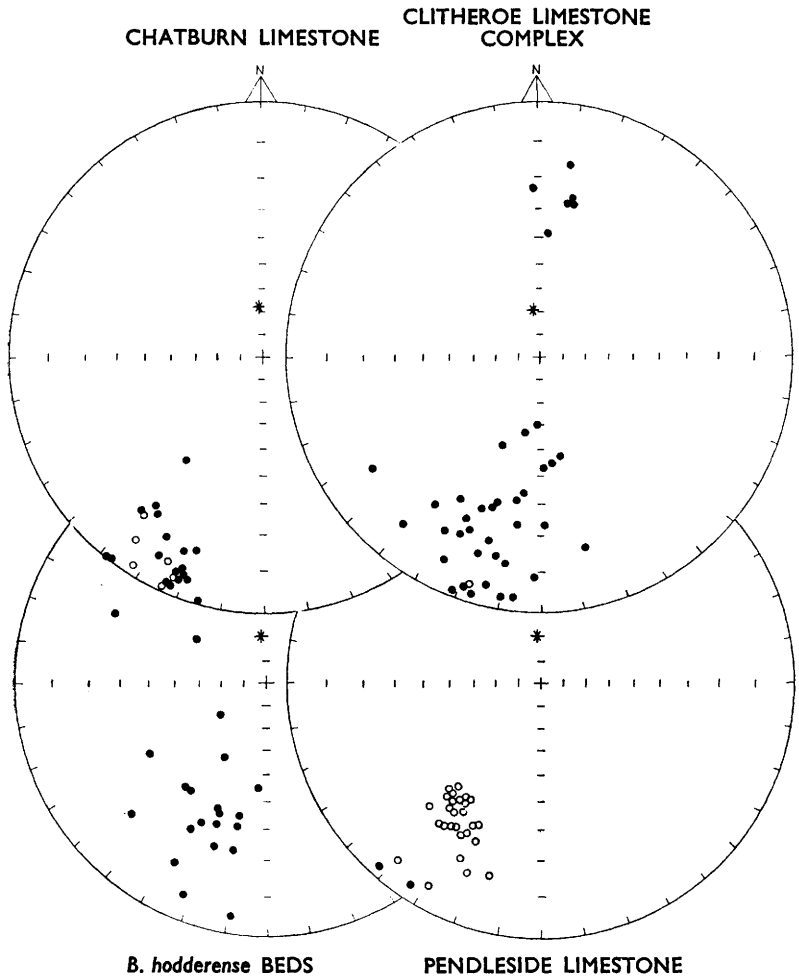


FIG. 2. Initial directions of NRM for Craven Basin Limestones. The projection is stereographic, circumference = horizontal, closed symbols = downward dip, open symbols = upward dip, and the directions are corrected for bedding tilt. The present Earth's field is shown as an asterisk.

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shallow positive inclination. The most stable directions of magnetization of the limestones from both areas usually lay in the range 100–400°C. In most cases, by 400°C only about 10–20% of the initial remanence remained and the thermal demagnetization curves of intensity are all very similar (Fig. 4). Susceptibility behaviour during thermal demagnetization was also uniform. Initially the specimens showed only a small positive or negative susceptibility ($-1{\cdot}4$ to $+13{\cdot}0 \times 10^{-6}$ gauss oersted^{-1}) but at about 400°C a large increase (up to 125×10^{-6} gauss oersted^{-1}) usually occurred indicating that important chemical changes took place at and above this temperature. Measurement of Curie point curves in air and *in vacuo* indicate that these chemical changes are the formation of magnetite and pyrrhotite due to the decomposition of pyrite.

On the basis of a visual inspection and computer analysis of the results from the pilot specimens all the remaining specimens were partially demagnetized at 200°C (Figs. 3*b* and 5, Tables 1, 3). The specimens from the Craven Basin are well grouped although there are differences in the mean vector directions for the four different limestone units. The mean directions of magnetization show a systematic stratigraphical variation from Dec: 205, Inc: −19, α_{95}: 6 (Chatburn Limestone); Dec: 199, Inc: −6, α_{95}: 13 (Clitheroe Limestone Complex); Dec: 192, Inc: −3, α_{95}: 11 (*B. hodderense* Beds) to Dec: 216, Inc: −31, α_{95}: 10 (Pendleside Limestone). The specimens of the purer Great Scar Limestone formed two groups after partial demagnetization (Fig. 3*b*). One group with SSW declination and a shallow negative inclination is similar to the directions found in the Craven Basin. Another group in the NE quadrant is very scattered with positive and negative inclinations. Piper (1973) measured a variety of facies from the Great Scar Limestone with similar results.

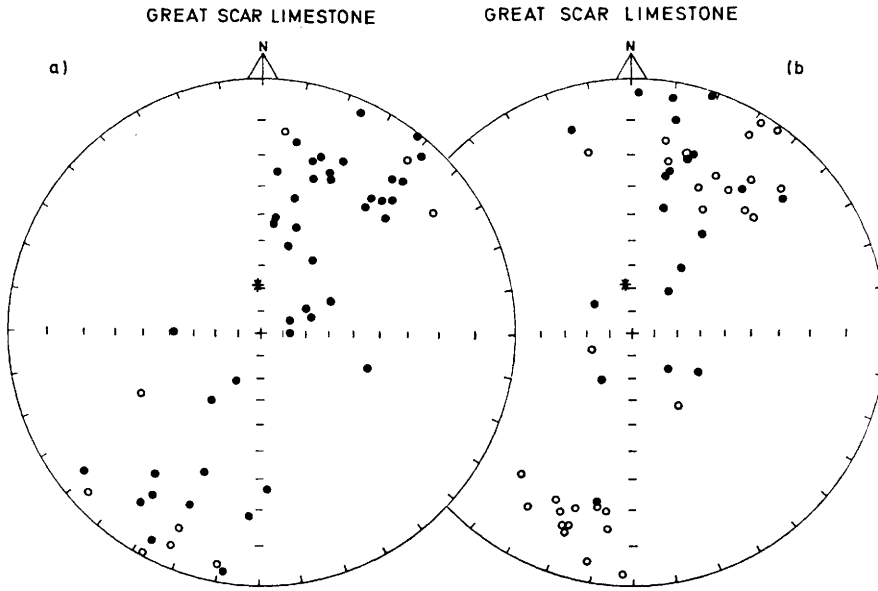


FIG. 3. Directions of NRM for the Askrigg Block (a) before and (b) after partial demagnetization at 200°C. Details of projection as in Fig. 2.

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Origin of the magnetization

The time of acquisition of natural remanent magnetization is fundamental and must be determined before any interpretations can be made. Both the initial and cleaned directions of NRM shown in Figs. 2, 3 and 5 refer to the present day horizontal. The thermal demagnetization curves (Fig. 4) show a large decrease in the intensity of remanence above 250°C and the sharp drop from 350–400°C suggests that this is near to the Curie point of the magnetic mineral carrying the NRM. The observed maximum blocking temperature is about 500°C in samples from the *B. hodderense* Beds. Alternating field demagnetization curves (Fig. 6) also show a large decrease in intensity in fields of only a few hundred oersteds and it thus seems likely that the remanence is carried principally by magnetite or titanomagnetite. Turner (1975) in a detailed study of the origin of the magnetization was able to confirm the presence of magnetite in some of the limestones. A test comparing the magnitude of the saturation anhysteritic remanent magnetization with the NRM suggested that many of the limestones have a very low net alignment of magnetic vectors. This is partial evidence that the remanence is of depositional rather than chemical origin. Moreover many of the limestones show a weak but well defined magnetic susceptibility anisotropy which lies mainly within the limestone bedding planes and it thus seems very likely that the magnetization is a depositional remanent magnetization (DRM).

AF demagnetization of the IRM and ARM shows that the ARM is less stable than the IRM and NRM. In all cases the stability increases with increasing strength of the inducing field. The test of Lowrie & Fuller (1971) for the presence of multidomain magnetite which was used by McElhinny & Opdyke (1973) in a study of the Trenton Limestone has been applied to these Carboniferous limestones. The test compares the stability of TRM (or ARM) and IRM to AF demagnetization. In a multidomain magnetite IRM is most stable. In the Carboniferous limestones studied here, both the NRM and IRM are much more stable than the ARM, indicating that the majority of the magnetite is in the multidomain state.

TABLE 2: *Most stable range and stability index of the pilot specimens*

Pilot Specimen	Most stable direction Dec.	Inc.	α_{95}	Most stable temperature Range	Stability Index
Craven Basin					
PL 1.2	226	-24	2	300-400	8.83
PL 3.2	205	-25	3	100-400	4.91
PL 4.1	224	-38	2	100-300	9.58
PL 5.2	191	-25	2	100-350	6.37
APH 2B	30	+38	2	100-200	7.76
APH 33	29	+20	33	20-250	0.55
APH 3C	41	+37	3	100-300	6.10
BH 1.1	191	-7	3	300-400	4.80
BH 3.2	184	-3	3	100-400	4.50
APH 6B.1	191	-15	4	200-300	3.90
APH 7B.2	8	+62	9	150-400	1.69
WS 3.3	188	+8	6	200-300	2.54
WS 4.4	9	+12	8	20-350	1.80
WS 5.4	206	-8	12	100-250	1.58
WS 6.2	207	-18	7	100-350	2.10
WS 7.4	200	+27	25	100-250	0.77
CH 1.1	211	-27	1	200-300	12.86
CH 3.2	216	-25	6	100-350	2.51
CH 4.1	208	-12	12	20-250	1.43
CH 5.2	193	-25	16	100-300	0.99
APH 4	204	-11	4	100-300	3.74
Askrigg Block					
APH 16A	17	+3	10	200-400	2.16
APH 15B	2	-3	10	150-300	1.95
APH 30	34	-15	12	100-350	1.50
APH 31	8	+3	31	100-350	0.61
APH 8B	13	+25	5	100-200	3.19
APH 9A	30	+42	30	20-500	0.59
APH 10C	206	-9	9	100-300	1.86
APH 11B	192	+25	13	150-300	1.40
APH 12B	221	-17	19	100-400	0.90
APH 13	285	+64	38	200-500	0.56
APH 14	37	+15	5	20-400	3.27

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Johnson *et al.* (1974) have shown that the ARM stability of fine particle magnetite (0·016–0·2 μ) decreases with increasing field strength whereas that of coarse particle magnetite (210–250 μ) increases with increasing field strength. This suggests that the bulk of the magnetite in the limestones is relatively coarse grained and multidomained.

It was found possible to separate the magnetic minerals from the limestones by crushing to −150 mesh and using bromoform in a centrifuge. Polished sections of the resultant extracts show a complex association of haematite, haematite-ilmenites, magnetites (including titanomagnetite) and pyrite. Many of the haematites and magnetites show secondary oxidation features and are clearly of detrital origin. The petrographic and magnetic evidence is thus consistent with the primary magnetization of the limestones being a DRM carried by magnetite. The preservation of magnetite in Palaeozoic sedimentary rocks is interesting since van Houten (1968) has shown that it readily converts to haematite with time. Its preservation is due to the fact that the early diagenetic environment of the more strongly magnetized limestones was quite strongly reducing. This is indicated by the presence of diagenetic pyrite, carbon fragments and bitumens. Berner (1970) showed that pyrite formed by the reduction of dissolved sulphate and Fe^{3+}, through the intermediate ferromagnetic mineral pyrrhotite, but there is no evidence of the presence of pyrrhotite in any of the limestones. The preservation of the DRM was ensured due to the carbonate cementation which followed the early geochemical changes fairly rapidly.

The time taken for the magnetization to be acquired by the limestone is difficult to estimate but because it is a DRM only the sedimentation time for a 2·5 cm interval (the thickness of the measured specimens) is needed. By dividing the maximum thickness of sediment in the Craven Basin by its radiometric age a figure of 185 yr/2·5 cm of compacted sediment is obtained. However, because of breaks

TABLE 3: *Site mean directions and intensities of the Great Scar Limestone in the Askrigg Block. N refers to the number of specimens in each hand sample; k is Fisher's (1953) precision parameter and α_{95} is the 95% cone of confidence for the mean.*

		Initial					200°C				
Site	N	Dec.	Inc.	Int.($\times10^{-6}$ gauss)	k	α_{95}	Dec.	Inc.	Int.($\times10^{-6}$ gauss)	k	α_{95}
Zone D_1											
APH 16	5	21	+23	0.14	41	12	13	-5	0.15	13	13
Zone S_2-D_1											
APH 15	5	75	+61	0.03	1	90	91	-33	0.04	1	90
APH 30	5	41	+5	0.19	63	10	41	-19	0.22	80	9
APH 31	6	29	+25	0.06	8	25	17	+17	0.06	5	33
Zone S_1-S_2											
APH 8	4	10	+33	0.27	83	10	15	+23	0.25	320	7
APH 9	5	72	+70	0.11	78	9	25	+15	0.07	5	39
APH 10	5	212	+1	0.25	49	11	200	-15	0.22	251	5
APH 11	5	193	+12	0.19	28	15	189	-6	0.20	21	17
APH 12	5	239	+23	0.10	1	90	237	-54	0.11	1	90
APH 13	4	220	+44	0.07	4	50	203	+66	0.04	1	90
APH 14	3	39	+17	0.47	213	8	43	+17	0.49	90	27

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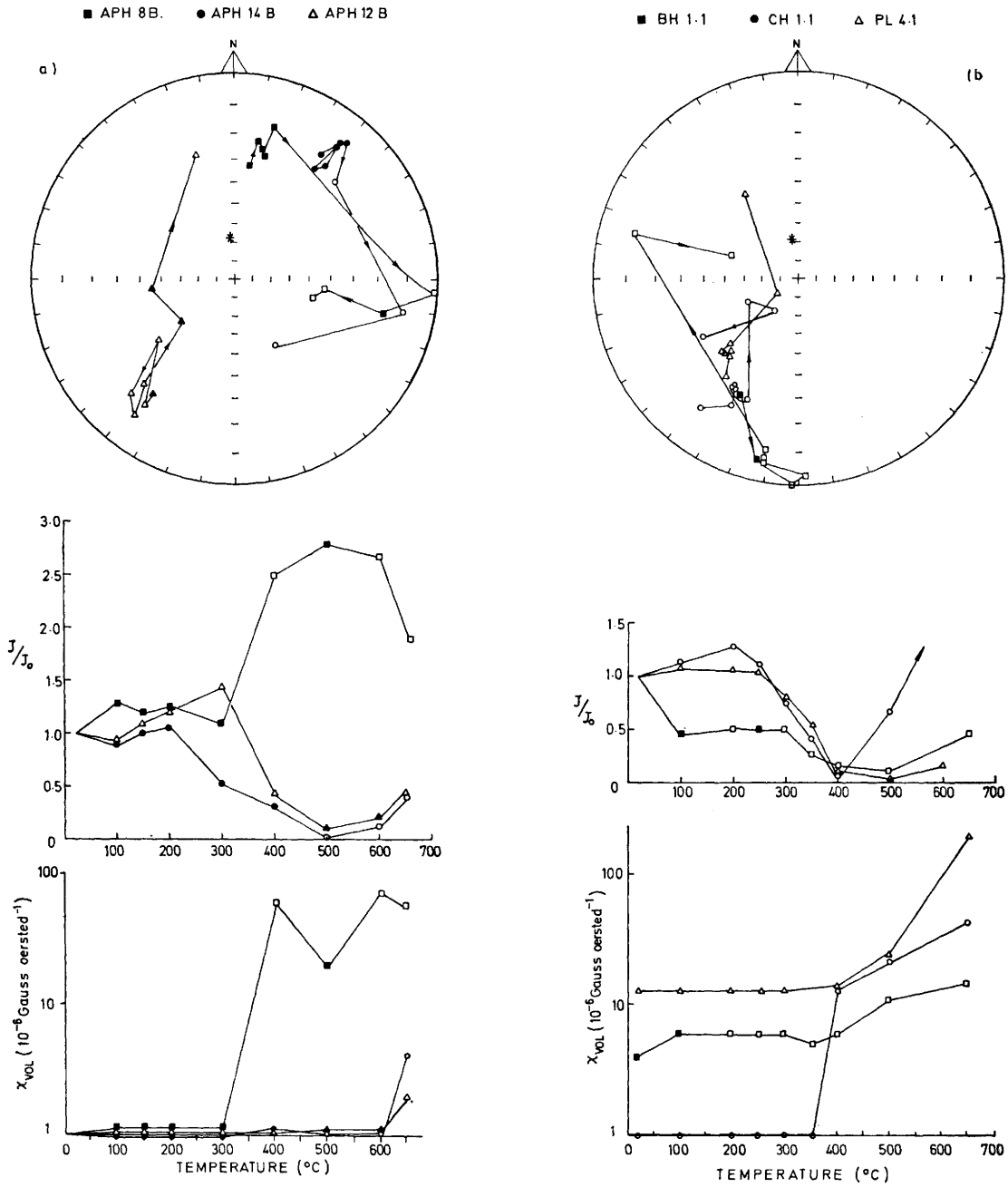


FIG. 4. Examples of the behaviour of pilot specimens in partial thermal demagnetization including directions of remanence, intensity (M/M_0) and susceptibility (χ_{vol}) (a) for the Askrigg Block (Great Scar Limestone) and (b) for the Craven Basin. Details of projection as in Fig. 2.

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in sedimentation and post-depositional compaction this must be regarded as an absolute maximum and a figure near 50 yr/2·5 cm is probably more accurate (Bahaman aragonite muds have accumulated at the rate of 32 yr/2·5 cm). These figures indicate that the limestones are potentially useful for studying Carboniferous secular variation.

Carboniferous polarity scale

A pre-Kiaman polarity scale (Fig. 7) has been constructed for the two countries which are now most fully documented, i.e. USSR and Great Britain. For the British section the scale was compiled from this work, Hicken *et al.* (1972) and the data of McAuley & Morris (1971). The main source of information for the USSR section was Khramov (1967, 1971).

Each zone has been designated R, R_m or M depending on the number of specimens measured being either all reversed, mostly reversed or about equally

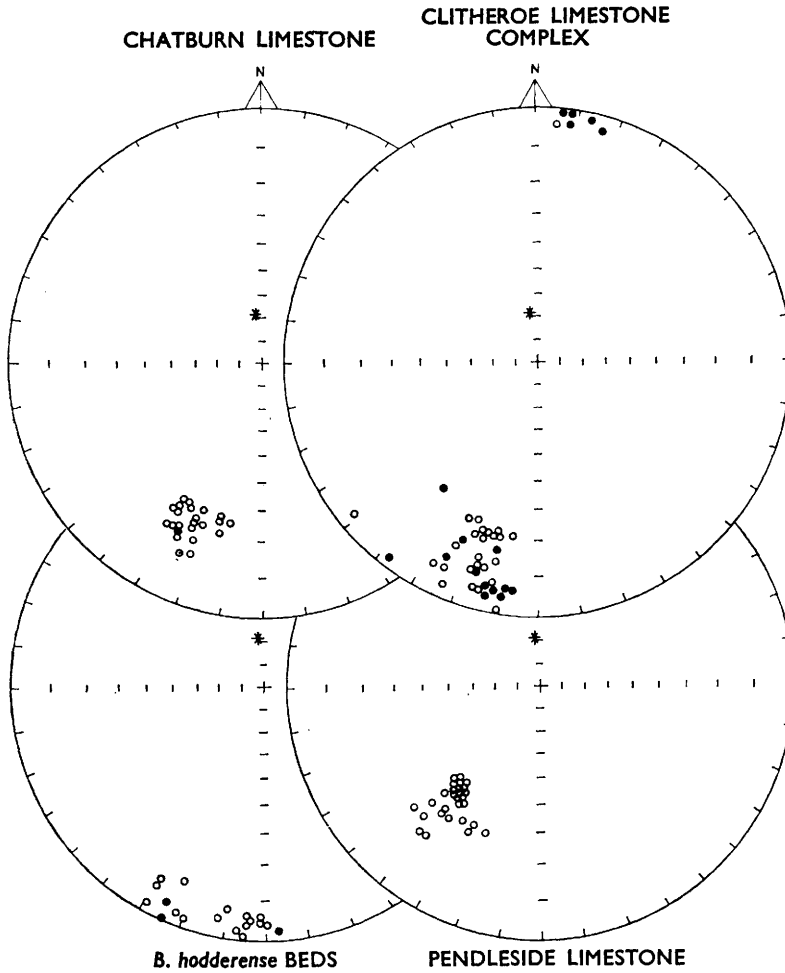


FIG. 5. Directions of remanence for limestones from the Craven Basin after partial demagnetization at 200°C. The details of the projection are the same as in Fig. 2.

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reversed and normally magnetized, respectively. There were apparently no periods during the lower Carboniferous when the geomagnetic field was normal for any length of time. In Britain there are no detected normal periods at all until upper C_2S_1 times. This agrees with the Russian data except that here there are some normal periods found in the Tournaisian. However, there are still very few data from the British Tournaisian. Mixed polarities dominate the middle and upper Viséan in Russia. The middle Viséan (upper C_2S_1–S_2) in Britain is also characterized by mixed polarities but there is a discrepancy in the upper Viséan (*D* zone) where there are less normal periods recorded in Britain than in the Russian section. This might be due to the difficulty of assigning a precise stratigraphical age to some of the Russian samples. The little evidence available suggests that the Namurian and immediately pre-Kiaman interval was dominated by mixed polarities.

The mid-Carboniferous polar 'transition'

The original polar wandering curves for North America and Europe (Collinson & Creer 1960) indicated a fairly steady average motion of the palaeomagnetic poles, with respect to each continent, of some 0·3° per million years. This would

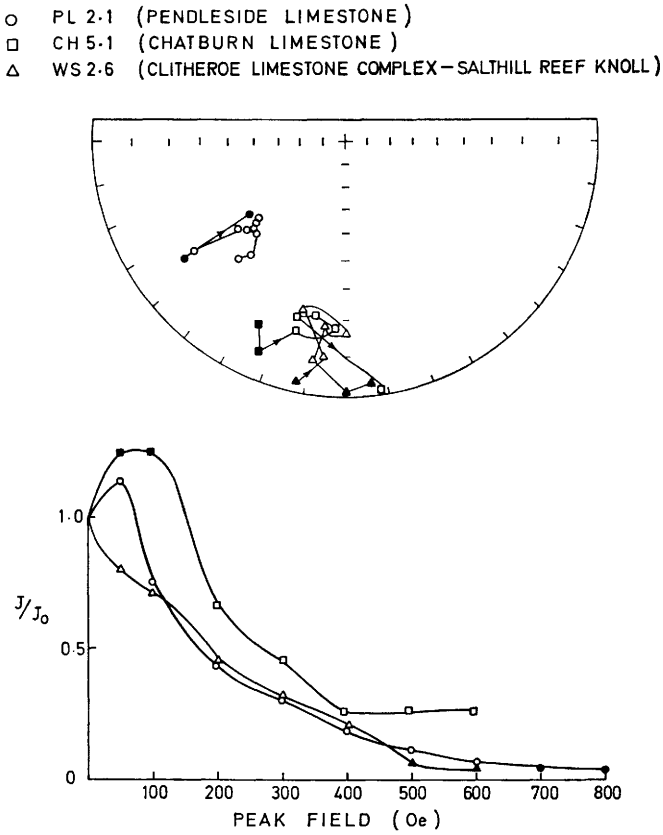


FIG. 6. Examples of the behaviour of the directions and intensity of pilot specimens to alternating field demagnetization.

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correspond to a palaeomagnetic polar movement of some 20° during the Carboniferous (345–285 m.y. B.P.) relative to all the continents. However, Irving (1966) demonstrated that all Australian upper Carboniferous to late Cretaceous palaeomagnetic poles were fairly tightly grouped and preceded by a movement of the palaeomagnetic pole relative to Australia by some 40° during the Namurian/Westphalian and also that there was some evidence for a 20° polar 'transition' within the European Carboniferous data. This concept of 'quasi-static' periods separated by brief polar 'transitions' was extended by Briden (1967) to all of the Gondwanan continents for which data were then available and McElhinny (1967) used evidence of a lower Carboniferous or possibly upper Devonian age for the Dwyka varves in S. Africa to determine a 55° polar movement relative to Africa some time during the Carboniferous, preceded by another 'transition' in the lower Devonian (McElhinny & Luck, 1970). In his review of palaeomagnetic data available up to early 1970, Creer (1970) concluded that a polar shift took place between the early and late Carboniferous in Europe and that data from other continents were consistent with that supposition. Tarling (1971) considered that palaeomagnetic data for the Palaeozoic were still too poorly defined for polar transitions to be dated although there was evidence for periods of slow continental motion relative to the pole. McElhinny (1973) no longer supported a mid-Carboniferous transition as an explanation for the African results, although he still retained a Gondwanan quasi-static interval in the late Carboniferous–Permian with the implication that this

STAGE	U.S.S.R.			GREAT BRITAIN		
NAMURIAN		Protvinian	M	Namurian		M
VISEAN	Serpukhovian	Steshevian and Tarussian	M	D	3	
					2	R
					1	R_M
	Okaian	Mikhailovkian and Aleksinian	R	S_2		M
	Yasnopolyanian	Tulian and Bobrikovkian	R	C_2S_1	Upper	R_M
					Lower	R
TOURNAISIAN	Cheryshinian	Kizelian and Cherepetian	R_M	C_1		R
	Likhvinian	Ozersko-Khovanian		Z		R
				K		?

FIG. 7. Polarity scale for the lower Carboniferous based on results from the USSR and Great Britain. R indicates that all specimens are reversed, R_M that most specimens (more than 75%) are reversed, and M that reversed and normally magnetized specimens are equally abundant.

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was preceded by a transition. Newer evidence relating to the postulated mid-Carboniferous polar transition can now be re-evaluated (Tables 4, 5).

The African data indicate a large movement, relative to the pole, between the Devonian and Triassic, but this could be an average motion of some 1° per million years which, although faster than the average Phanerozoic rate, is much less than the maximum known sea-floor spreading rate of some 3° per million years which

NORTH AMERICA		Pole Position			EUROPEAN		Pole Position		
		N	E				N	E	
Upper					Upper				
11-76	Cumberland Group	36	125	Roy (1969)	2-36	Whin Sill, UK	37	169	Creer et al(1959)
8-101	Pictou Group	41	132	Roy (1966)	11-77	Whin Sill, UK	44	159	Storetvedt et al(1969)
9-98	Hurley Creek Fmtn.	39	125	Roy et al (1968)	-	Holy Island Dyke,UK	38	177	Giddings et al(1971)
10-105	Maroon Fmtn.	33	126	McMahon & Strangway(1968)	-	Wackerfield Dyke,UK	22	175	Tarling et al(1973)
-	Upper Maroon Fmtn.	42	119	Christensen & Helsley(1974)	-	Exeter Lavas, UK	48	163	Cornwell (1967)
10-106	Fountains & Lyons Fmtn.	48	119	McMahon & Strangway(1968)	-	Exeter Sediments,UK	46	165	Cornwell (1967)
					-	Midland Valley,UK	41	190	Milton (1972)
-	Rico Fmtn.	44	120	Gose & Helsley(1971)	11-71	L.Silesia Volcanics	54	173	Birkenmajer et al(1968)
					10-111	Inner Sudetic	39	177	Birkenmajer et al(1968)
					10-112	Plzen	31	152	Birkenmajer et al(1968)
					10-113	Klavo-Rakovink	36	154	Birkenmajer et al(1968)
Middle-Upper					10-107	Mt Billinger Sill	31	174	Priem et al (1968)
10-115	Minturn Fmtn.	39	105	McMahon & Strangway(1968)	10-108	Mt Hunneberg Sill	38	177	Priem et al (1968)
-	M-U Carboniferous	52	110	Scott & Helsley(1973)	10-109	Skane Dol	37	174	Priem et al (1968)
-	Lower Maroon Fmtn.	41	133	Christensen & Helsley(1974)	-	Swedish sills	34	170	Mulder (1971)
					12-119	Ny Hellusund	39	161	Halvorsen (1970)
Middle					Middle-Upper				
-	Hopewell Sandstones	37	122	Roy & Park (1974)	-	Derbyshire, UK	35	116	Titman (1971)
11-78	Hopewell Group	34	118	Roy & Park (1969)	-	Millersdale, UK	12	160	Titman (1971)
					-	Ashford, UK	-7	199	Titman (1971)
Lower-Middle					Middle				
10-119	Maun Chunk Fmtn.	43	127	Knowles & Opdyke(1968)	-	Halvdanpiggen,Spitzb.	40	180	Halvorsen (1972)
10-120	Maringouin Fmtn.	34	117	Roy & Robertson(1966)					
					Lower				
Lower					8-118	Lavas, Scotland	18	161	Wilson & Everitt (1963)
					-	Bristol, UK	32	158	Morris et al (1973)
-	St Joe Limestone	36	130	Scott & Helsley(1973)	-	Irish Limestones	43	161	McAuley & Morris (1971)
10-126	Perry Fmtn.	32	118	Robertson et al(1958)	-	Pendleside, UK	44	127	This paper
					-	B.hodderense Beds,UK	37	163	This paper
					-	Clitheroe, UK	37	154	This paper
					-	Chatburn, UK	41	143	This paper

TABLE 4: *Most reliable European and North American Carboniferous data from non-tectonic regions. Hyphenated numbers refer to the Geophysical Journal pole lists.*

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occurred during the early evolution of the Indian Ocean (McKenzie & Sclater 1971). The South American evidence does not indicate any polar transition, but the dating of the pre-Kiaman Carboniferous is somewhat uncertain and the Devonian direction is not yet adequately defined. Similarly, the Indian data are too sparse for evaluation of mid-Carboniferous polar behaviour. This means that the

	POLE POSITIONS N	E	α_{95}	ANGULAR SEPARATION	REFERENCES	NOTES
Australia						
Kiaman'	45	313	8	> 28 >44	Irving 1966	1. Kiaman is the reversed polarity interval between
Basal Kiaman'	73	327	-	> 19 >37 >61	2, Irving 1966	U.Westphalian and uppermost Permian (Irving &
L.Carboniferous	73	34	-	> 36	Luck, 1973	Parry, 1963)
Devonian	69	175	30		Luck, 1973	2. Paterson Toscanite = basal Kiaman at 298 m.y.
Africa						
Triassic	69	255	8	> 52 >90	3	3. Permian data from the K3 Galula and Songwe-
Carboniferous	26	206	-	> 45	McElhinny & Opdyke (1968)	Ketewaka (Opdyke 1964) are widely separated and
M. Devonian	-17	220	-		Hailwood & Tarling (1973)	newer data from Algeria (Abou-Deeb, unpublished)
S. American						are intermediate between these. This could mean
Kiaman'	74	167	8	> 9 >66	Thompson (1972)	rapid motion during the Permian but the
Pre Kiaman'	65	164	12	> 58	Thompson (1972)	stratigraphic order is not clear.
Devonian	12	130	90		4	4. The two Devonian poles are poorly defined (α_{95} =
India						24 and 36) and 67° apart so that the mean position
Permo Trias	-8	122	15	> 42	Wensink (1968)	has little reliability.
U. Carboniferous	32	134	-		5	5. Table 6.
Europe ('Western' selected data)						6. 91 poles for Europe, West of the Urals, have been
U. Carboniferous	40	169	5	> 17 >13		published. Removal of data of uncertain age,
M.+M-U.Carboniferous	23	166	49	> 17		magnetic or tectonic stability leaves 26 poles
L. Carboniferous	37	153	10		6	determination based on data after the application
Europe ('Russian' data)						of 'standard' Western demagnetisation techniques
U. Carboniferous	39	178	6	> 11		at 26 based on standard 'Russian' techniques
M. Carboniferous	30	172	8	> 15 >20		(Tarling, 1971). Asian data is almost entirely
L. Carboniferous	33	155	10		6	of 'Russian' derivation.
Asia (E of 60°E) excl. Japan)						
U.+ U-M Carboniferous	56	174	17	> 23 >15		
M. Carboniferous	35	160	60	> 9		
L.+ L-M Carboniferous	44	196	21		5	
N. America						
U. Carboniferous	41	124	5	> 6 >8		
M. Carboniferous	36	120	10	> 2		
L. Carboniferous	34	119	10			

TABLE 5: *Mid Palaeozoic polar 'transitions'.*

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test for such a transition, on a world-scale, depends on European, North American and Asian data. These show anomalous polar determinations during the mid-Carboniferous, although it is not always clear to which part of the Carboniferous these dates actually apply. The better defined lower Carboniferous data now show a movement relative to these continents that is less than the average Phanerozoic rate. As a polar transition would be a world-wide event, the Australian observations must reflect either the polar excursion (or excursions) some time in the mid-Carboniferous or, more likely, tectonic events associated with the evolution of the Tasman geosyncline. In contrast, the evolution of the Uralian Ocean does not seem to have changed the slow rate of movement of the Laurasian continents relative to the pole, although it does appear to have changed the direction of motion.

Palaeogeography and palaeolatitudes

The most reliable data are available from North America and Europe, with considerable data also from Siberia. Palaeolatitude determinations for these three continental blocks may therefore be considered reliable, probably better than $\pm 5°$ for Laurasia and $\pm 10°$ for Siberia. On this basis, the data fully confirm the general relationships of these continents indicated by geometrical considerations (Bullard *et al.* 1965, Bott & Watts 1971). This agreement in the upper Carboniferous would be greater if Greenland were pivoted relative to North America, although this reconstruction has merely assumed transcurrent motion along the Nares Strait. One implication of this discrepancy is that tectonic movements in the Sverdrup Basin must have involved some anti-clockwise rotational movement of Greenland relative to North America prior to movement along the Nares Straits. Naturally the palaeolatitudes shown here are only strictly applicable over the region from which samples have so far been obtained (Figs. 8, 9) and major errors would be involved if such palaeolatitudes were extrapolated into areas (e.g. Mediterranean and Alaska) in which significant post Carboniferous movements have taken place relative to the main stable continental blocks.

During the Carboniferous there was a gradual clockwise rotation of the Laurasian block, but most of the land-mass remained in equatorial to tropical latitudes. Conversely, Siberia was moving from tropical–medium latitudes into high latitudes with the implication that the structures developed as the Uralian Ocean closed would contain a significant component of strike-slip motion. On their face value, the palaeomagnetic data from Siberia do not show a complete closure of the Uralian Ocean in the upper Carboniferous, but the angular discrepancy is within the errors of the data currently available, particularly from Siberia. It is significant, however, that parts of northern Siberia were already in very high latitudes in the upper Carboniferous, thereby reaching latitudes which, in the Permian, could result in the glaciation of elevated land, or possible minor ice-caps. Similarly this palaeogeography would mean that the Angaran flora, with its high latitude characteristics (Chaloner & Meyen 1973), was more likely to have been separated from the Euramerian flora by climatic zonation rather than by the distribution of land and sea, although its separation from the Cathaysian flora prob-

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ably involved both climatic and geographic separation until there was some mixing with Eastern European flora in the late Permian.

Data from the other continents are sparse (Table 5), with only one pole position determined for Africa and India for the whole Carboniferous, and there being some uncertainty about the precise age of both the African and some South American data (discussed above). Comparison of the African position with those of the better defined Laurasian positions indicates marginally better agreement for lower Carboniferous times, tending to confirm the probability of the Dwyka varves being of lower or middle Carboniferous rather than upper. Nonetheless, any comparison (Fig. 9) between India, Australia and the northern continents (or Africa) suggests that either different ages of rock were involved in sampling these areas or that the magnetization does not reflect the Carboniferous geomagnetic field adequately, i.e. that secular variations have not been averaged or that local tectonic movements render single observations inapplicable to the continental block as a whole. The South American data, which are greater in quantity, indicate a more equatorial position for South America than for Africa. However,

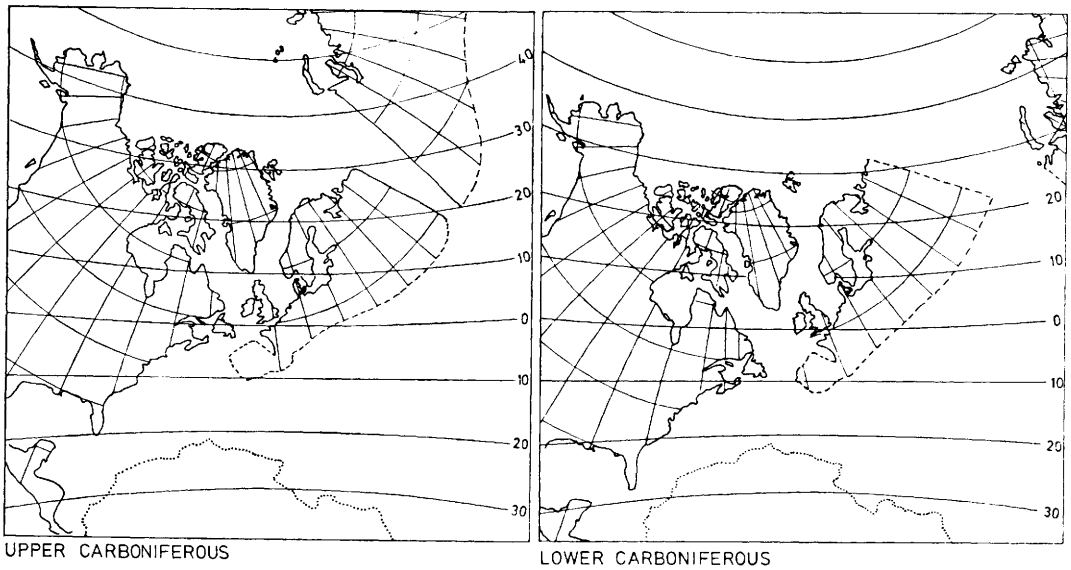
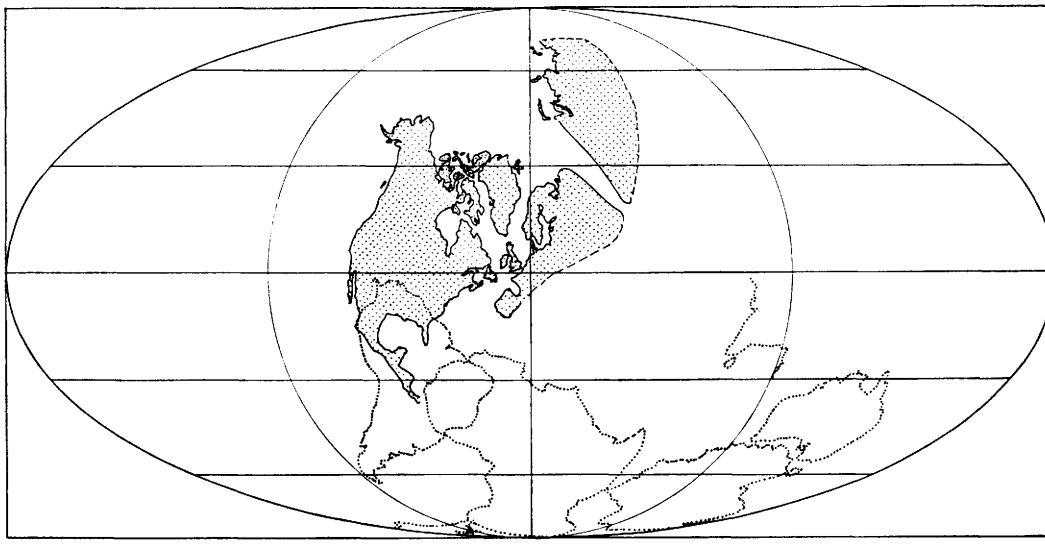


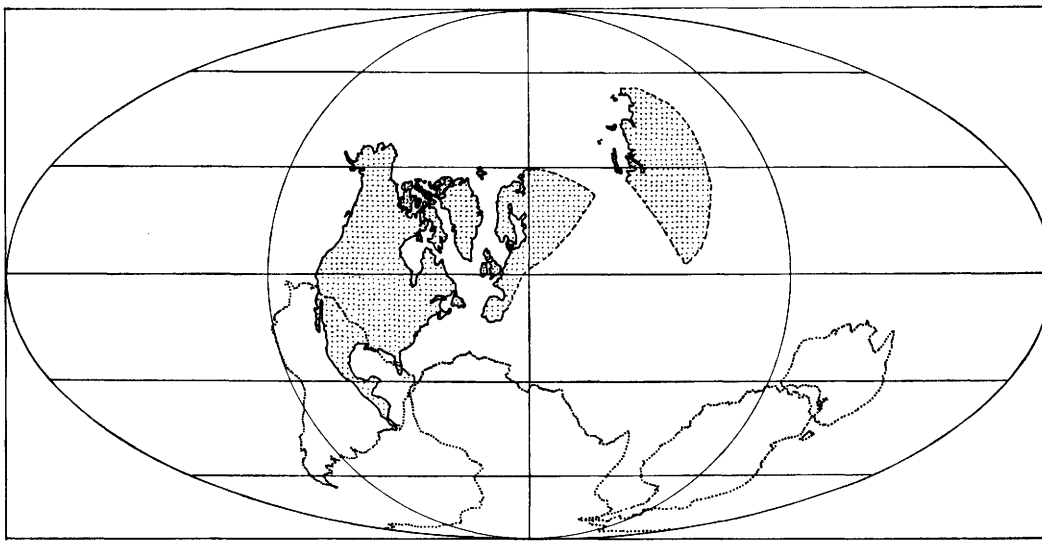
FIG. 8. Palaeogeography of the North Atlantic region. Both reconstructions are based on palaeomagnetic data (Table 5) to define the palaeolatitude of the continental blocks, but the longitudinal relationship is arbitrary. Greenland has been moved as a part of the North American continent after a small transcurrent adjustment along the Nares Straits. The southern European Boundary is for Alpine Europe in view of uncertainties about the location of this boundary in the Carboniferous period. The palaeolatitude definition for North America and Europe is of the order of 5° and for Siberia some 10°, but there is some uncertainty about the reliability of the African position (see text) which should be contiguous with the eastern North American coast at this period. The projection is stereographic and no adjustments have been made for within-continental movements since Carboniferous time, i.e. Alaska is shown for its present position relative to North America as its previous position is still uncertain.

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the position of South America relative to North America is only inconsistent by some 10–15° of latitude, which must be considered within the limits of the reliability of the South American data. This implies a more equatorial position for Africa, as required if the eastern seaboard of North America were then contiguous with the NW African coast.



UPPER CARBONIFEROUS



LOWER CARBONIFEROUS

FIG. 9. Palaeomagnetic reconstructions for the Carboniferous. The positions of the North Atlantic continents and Siberia are as in Fig 8. The other continents for which palaeomagnetic data are available are shown, with Antarctica assumed to be linked to Australia. The precision of the data for these other continents is low and the ages often uncertain (see text).

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